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Supply, Demand, and Government Policies



Supply, Demand, and Government Policies

- In a free, unregulated market system, market forces establish equilibrium prices and exchange quantities.
- While equilibrium conditions may be efficient, it may be true that not everyone is satisfied.
- One of the roles of economists is to use their theories to assist in the development of policies.



CONTROLS ON PRICES

- Are usually enacted when policymakers believe the market price is unfair to buyers or sellers.
- Result in government-created price ceilings and floors.



CONTROLS ON PRICES

- *Price Ceiling*

- A legal *maximum* on the price at which a good can be sold.

- *Price Floor*

- A legal *minimum* on the price at which a good can be sold.

How Price Ceilings Affect Market Outcomes

- Two outcomes are possible when the government imposes a price ceiling:
 - The price ceiling is not binding if set above the equilibrium price.
 - The price ceiling is binding if set below the equilibrium price, leading to a shortage.

Figure 1 A Market with a Price Ceiling

(a) A Price Ceiling That Is Not Binding

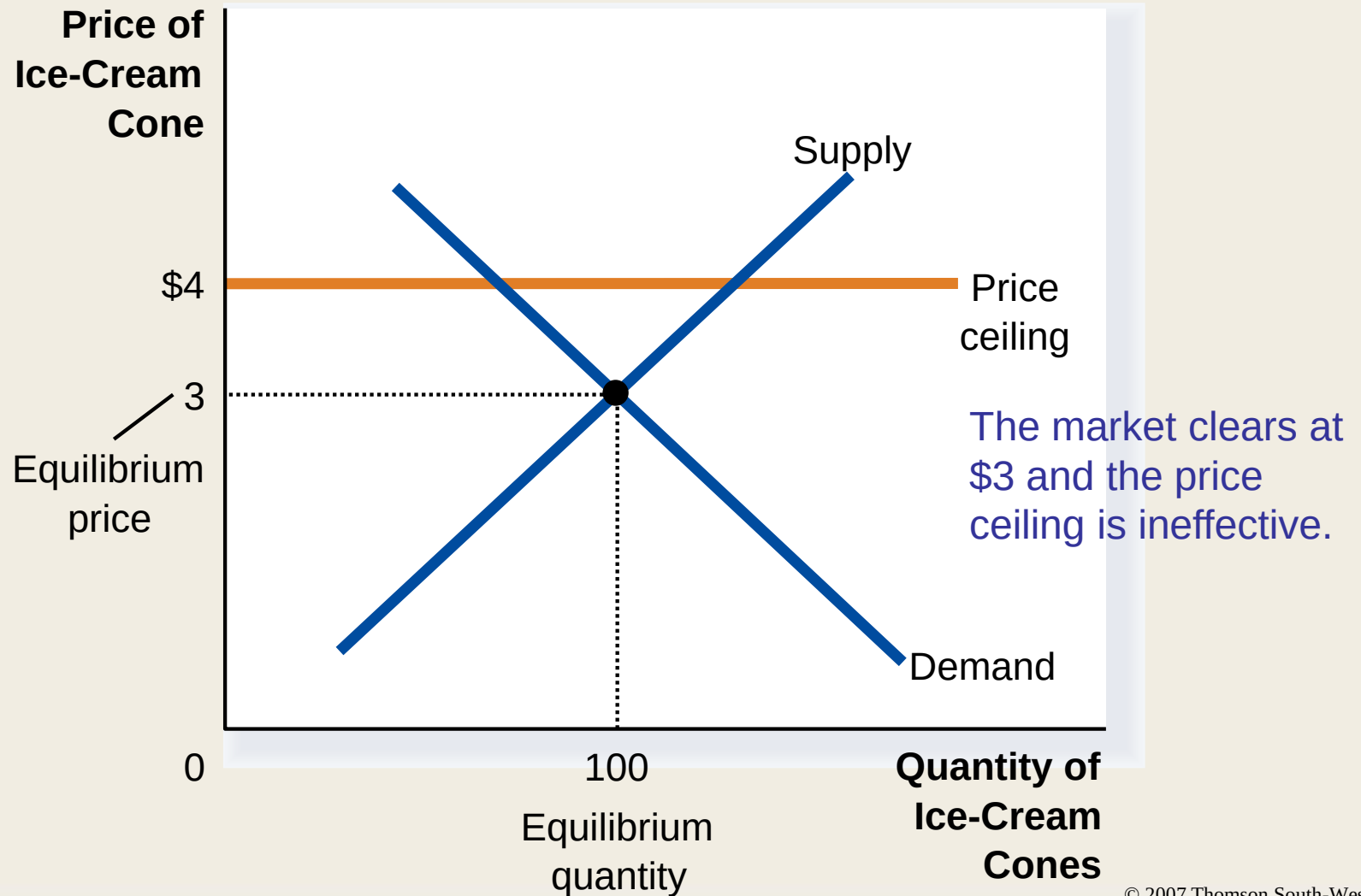
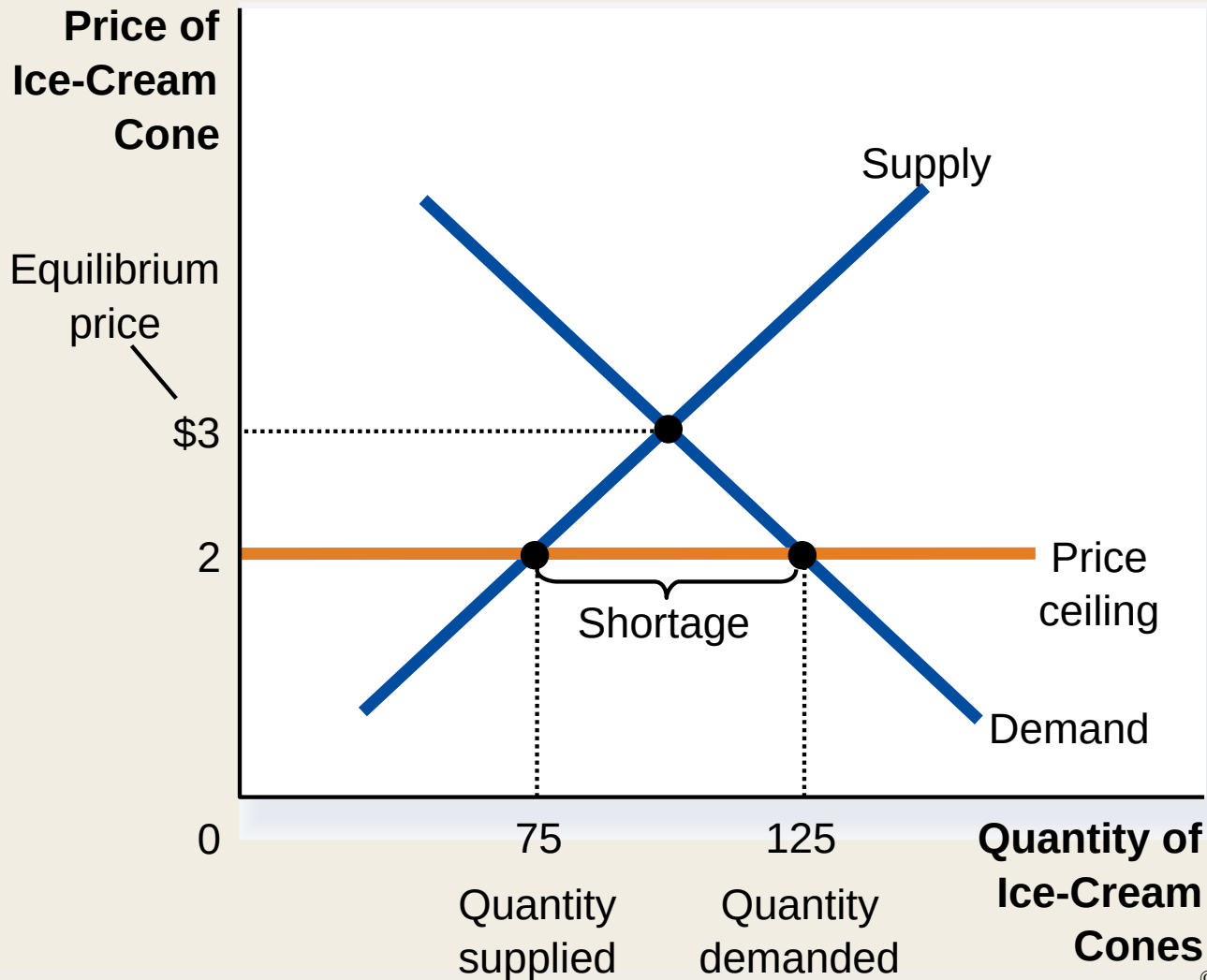


Figure 1 A Market with a Price Ceiling

(b) A Price Ceiling That Is Binding



How Price Ceilings Affect Market Outcomes

- Effects of Price Ceilings
- A binding price ceiling creates
 - Shortages because $Q_D > Q_S$.
 - Example: Gasoline shortage of the 1970s
 - Nonprice rationing
 - Examples: Long lines, discrimination by sellers

CASE STUDY: Rent Control in the Short Run and Long Run

- Rent controls are ceilings placed on the rents that landlords may charge their tenants.
- The goal of rent control policy is to help the poor by making housing more affordable.

Figure 3 Rent Control in the Short Run and in the Long Run

(a) Rent Control in the Short Run
(supply and demand are inelastic)

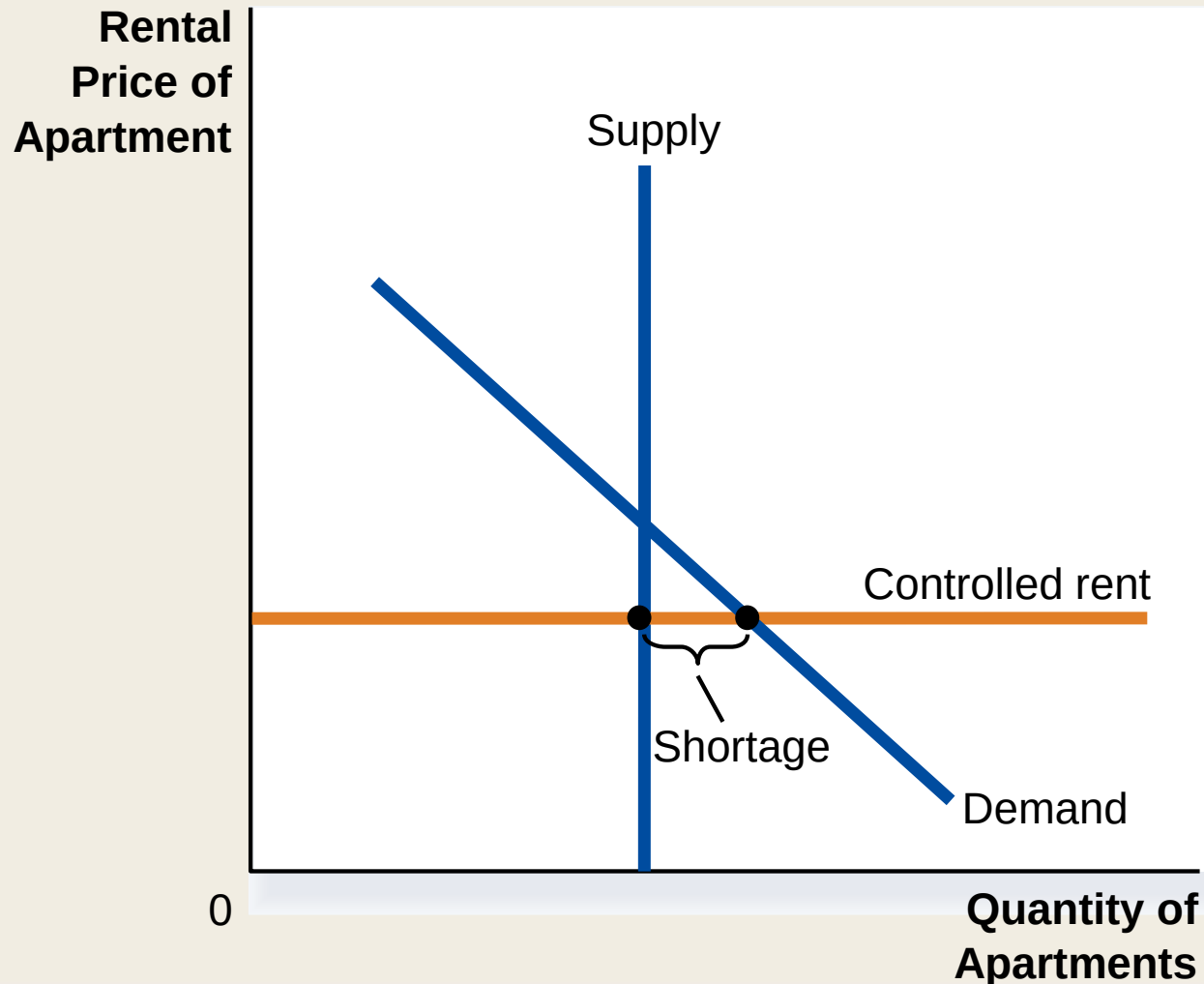
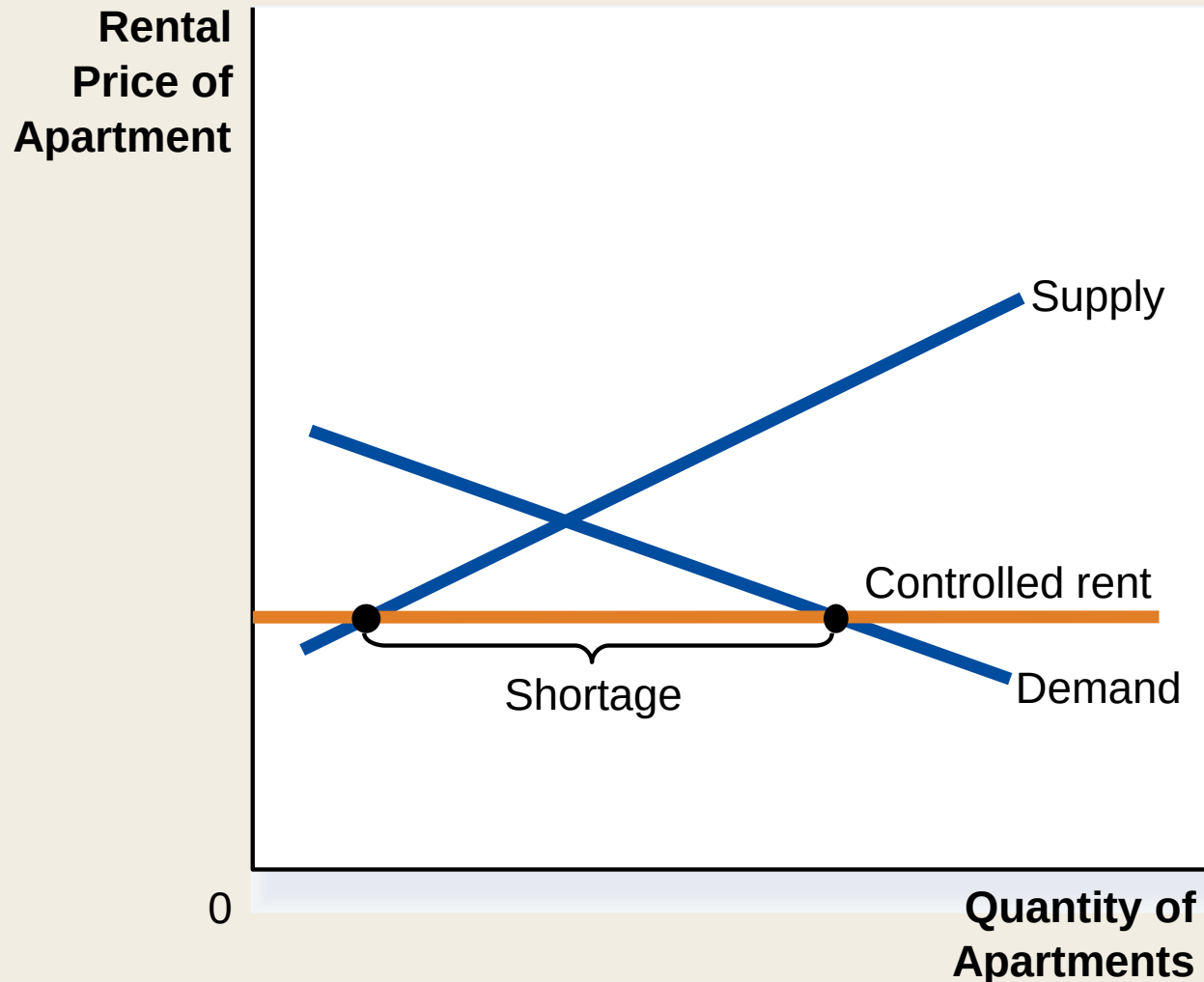


Figure 3 Rent Control in the Short Run and in the Long Run

(b) Rent Control in the Long Run
(supply and demand are elastic)

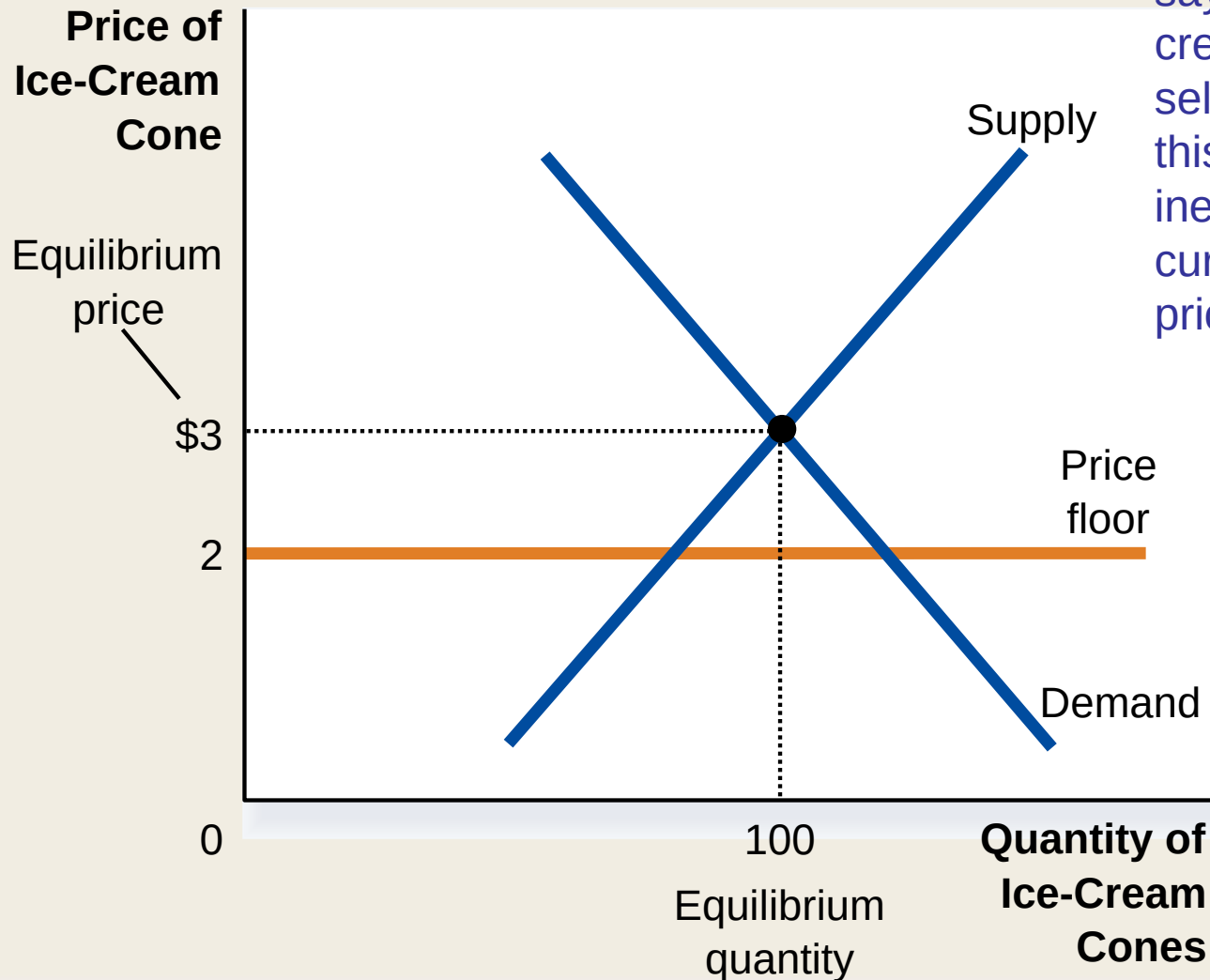


How Price Floors Affect Market Outcomes

- When the government imposes a price floor, two outcomes are possible.
 - The price floor is not binding if set below the equilibrium price.
 - The price floor is binding if set above the equilibrium price, leading to a surplus.

Figure 4 A Market with a Price Floor

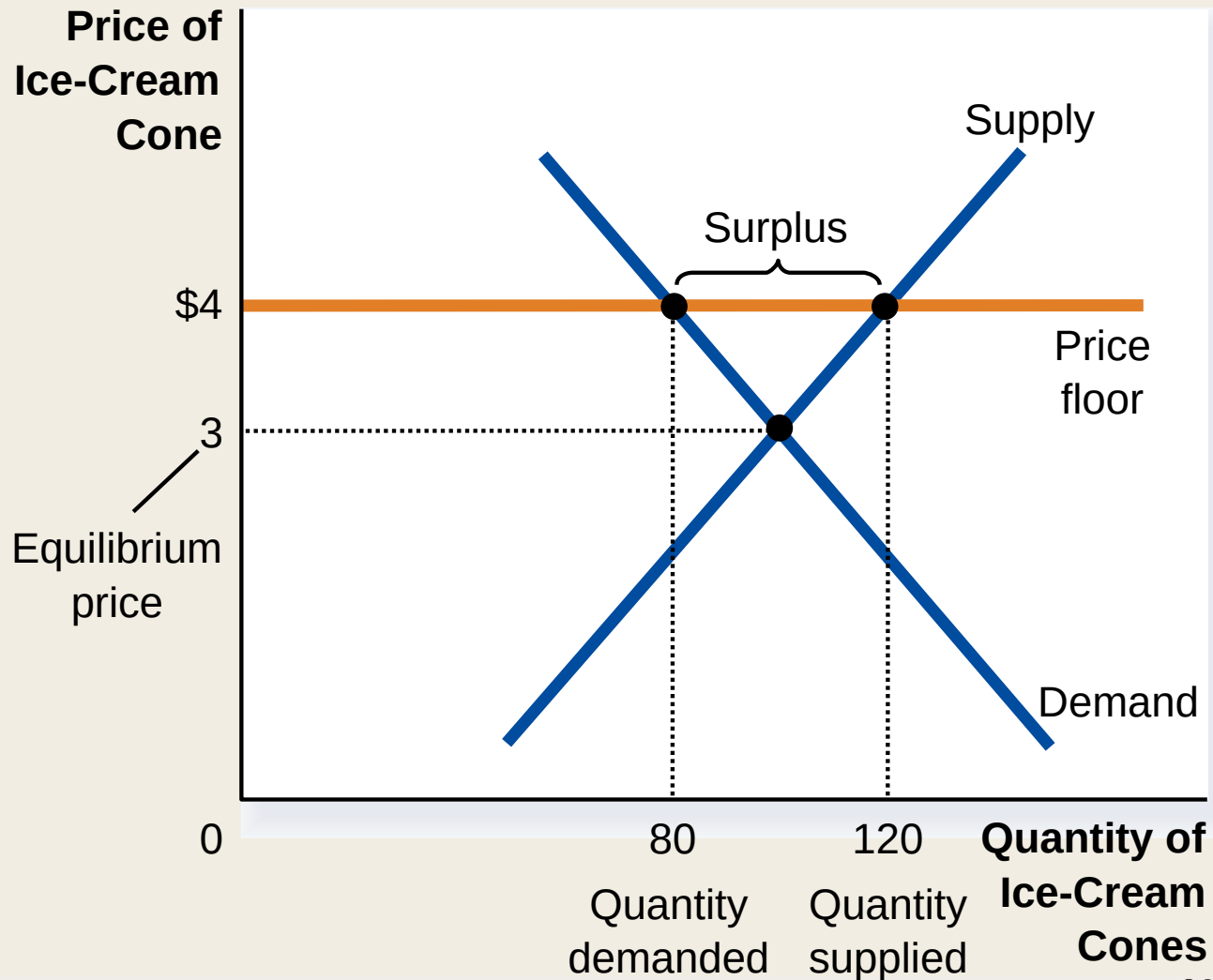
(a) A Price Floor That Is Not Binding



The government says that ice-cream cones must sell for at least \$2; this legislation is ineffective at the current market price.

Figure 4 A Market with a Price Floor

(b) A Price Floor That Is Binding



How Price Floors Affect Market Outcomes

- A price floor prevents supply and demand from moving toward the equilibrium price and quantity.
- When the market price hits the floor, it can fall no further, and the market price equals the floor price.

CASE STUDY: The Minimum Wage

- An important example of a price floor is the minimum wage.
- Minimum wage laws dictate the lowest price possible for labor that any employer may pay.

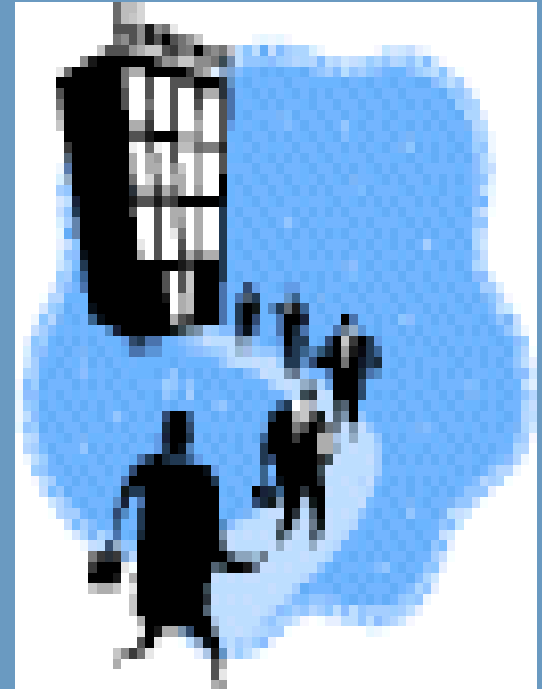


Figure 5 How the Minimum Wage Affects the Labor Market

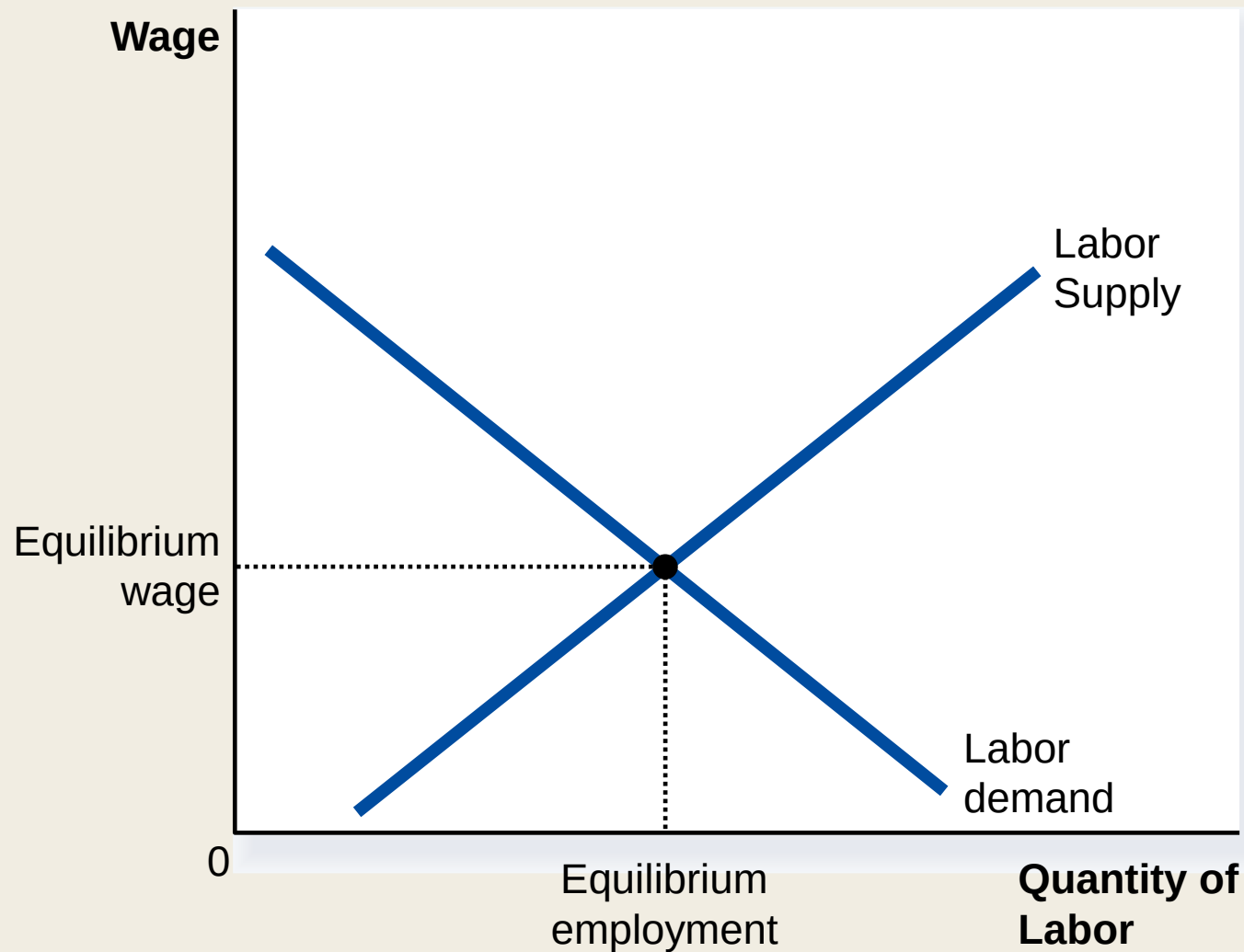
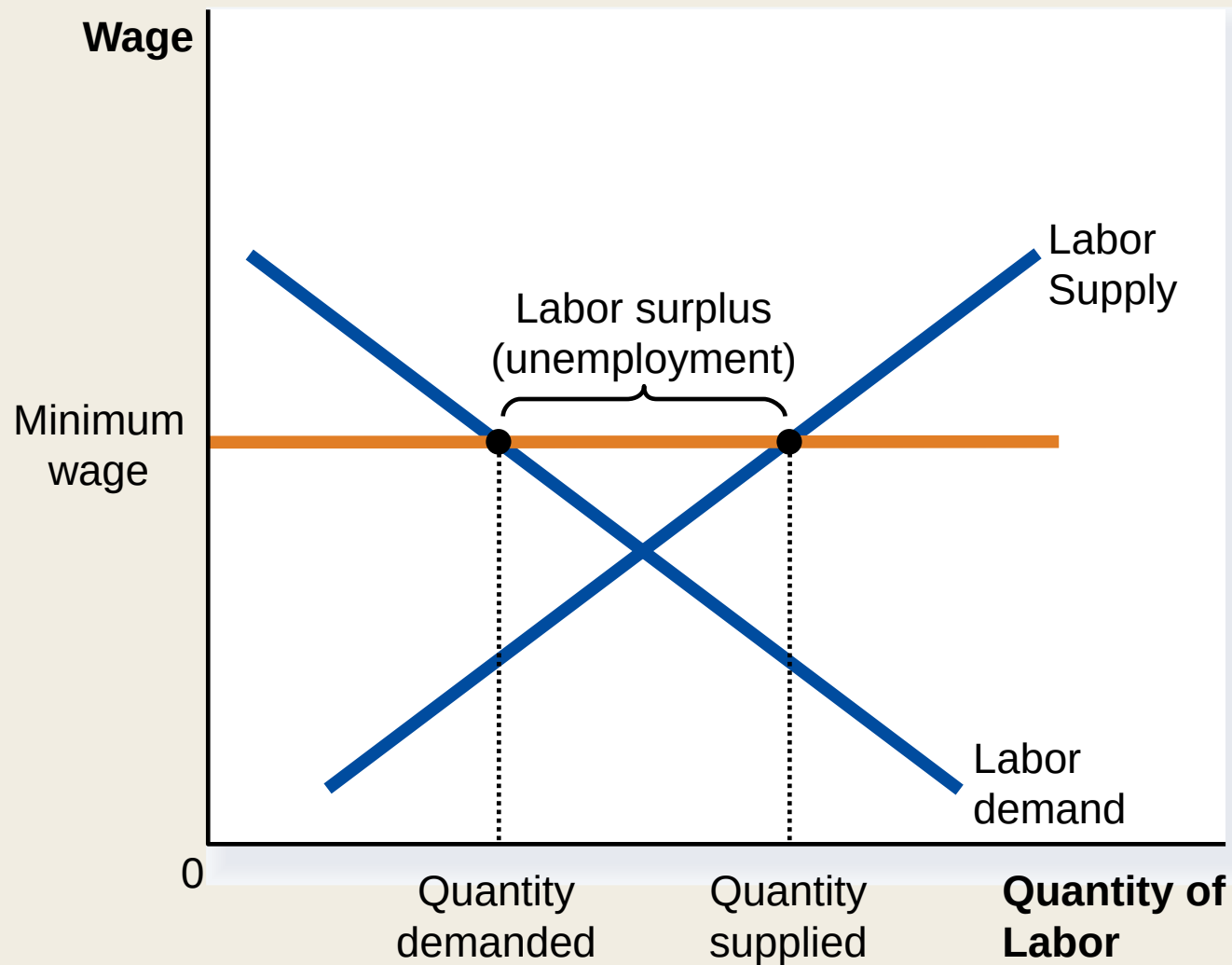


Figure 5 How the Minimum Wage Affects the Labor Market





TAXES

- Governments levy taxes to raise revenue for public projects.

How Taxes on Buyers (and Sellers) Affect Market Outcomes

- Taxes discourage market activity.
- When a good is taxed, the quantity sold is smaller.
- Buyers and sellers share the tax burden.



How Taxes on Buyers Affect Market Outcomes

- Elasticity and tax incidence
 - *Tax incidence* is the manner in which the burden of a tax is shared among participants in a market.

How Taxes on Buyers Affect Market Outcomes

- Elasticity and Tax Incidence
 - Tax incidence is the study of who bears the burden of a tax.
 - Taxes result in a change in market equilibrium.
 - Buyers pay more and sellers receive less, regardless of whom the tax is levied on.

Figure 6 A Tax on Buyers

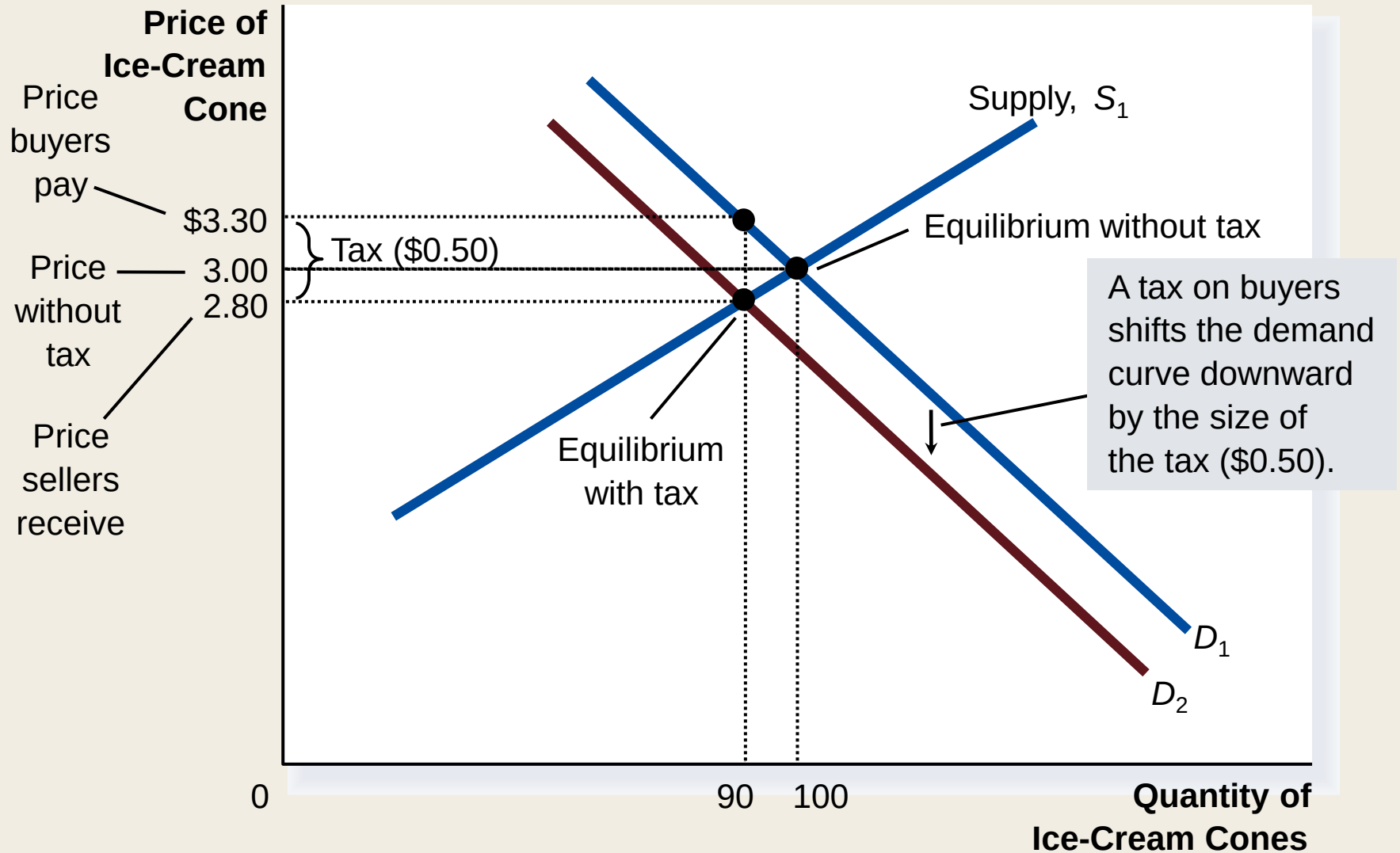
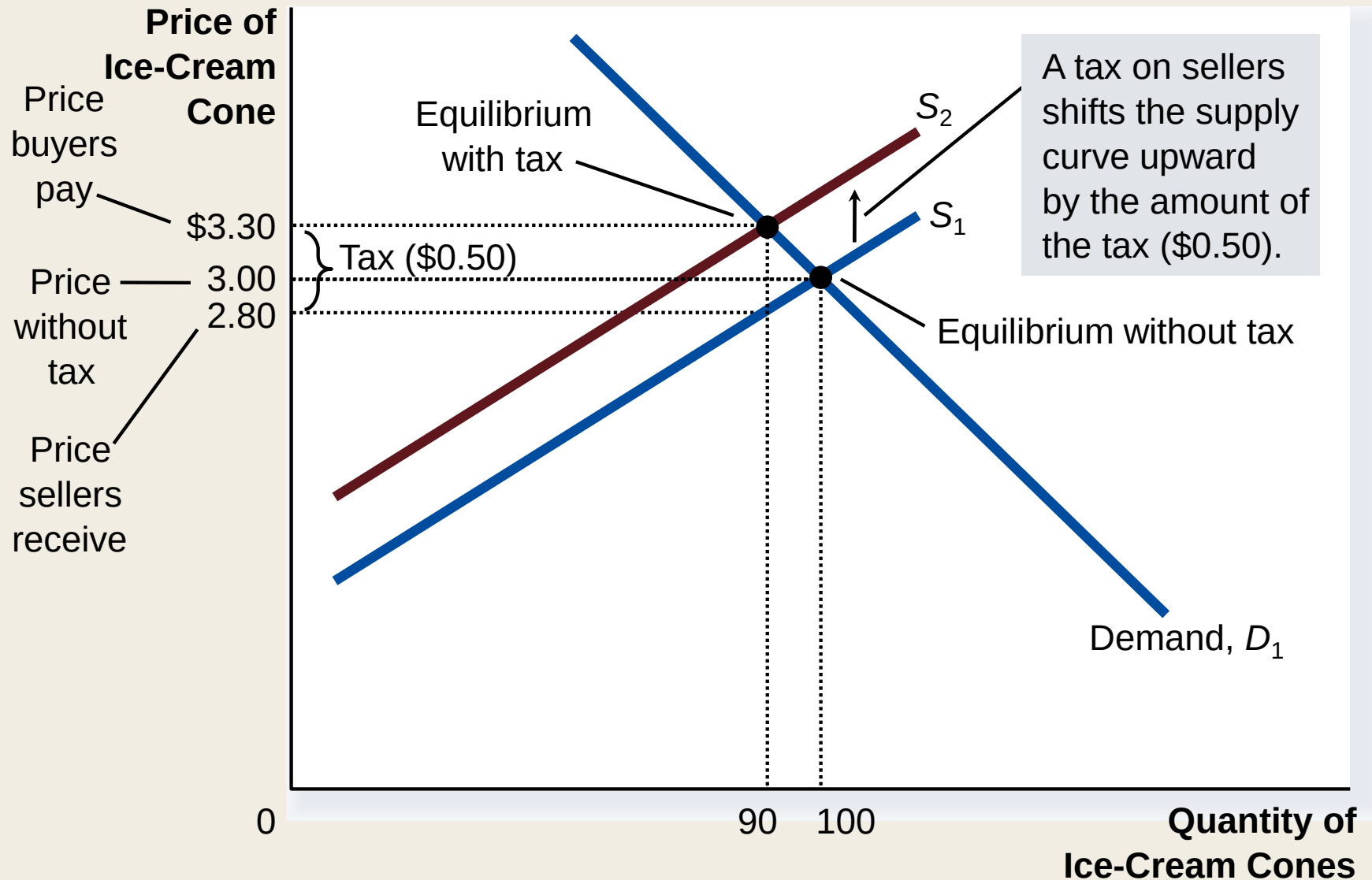


Figure 7 A Tax on Sellers

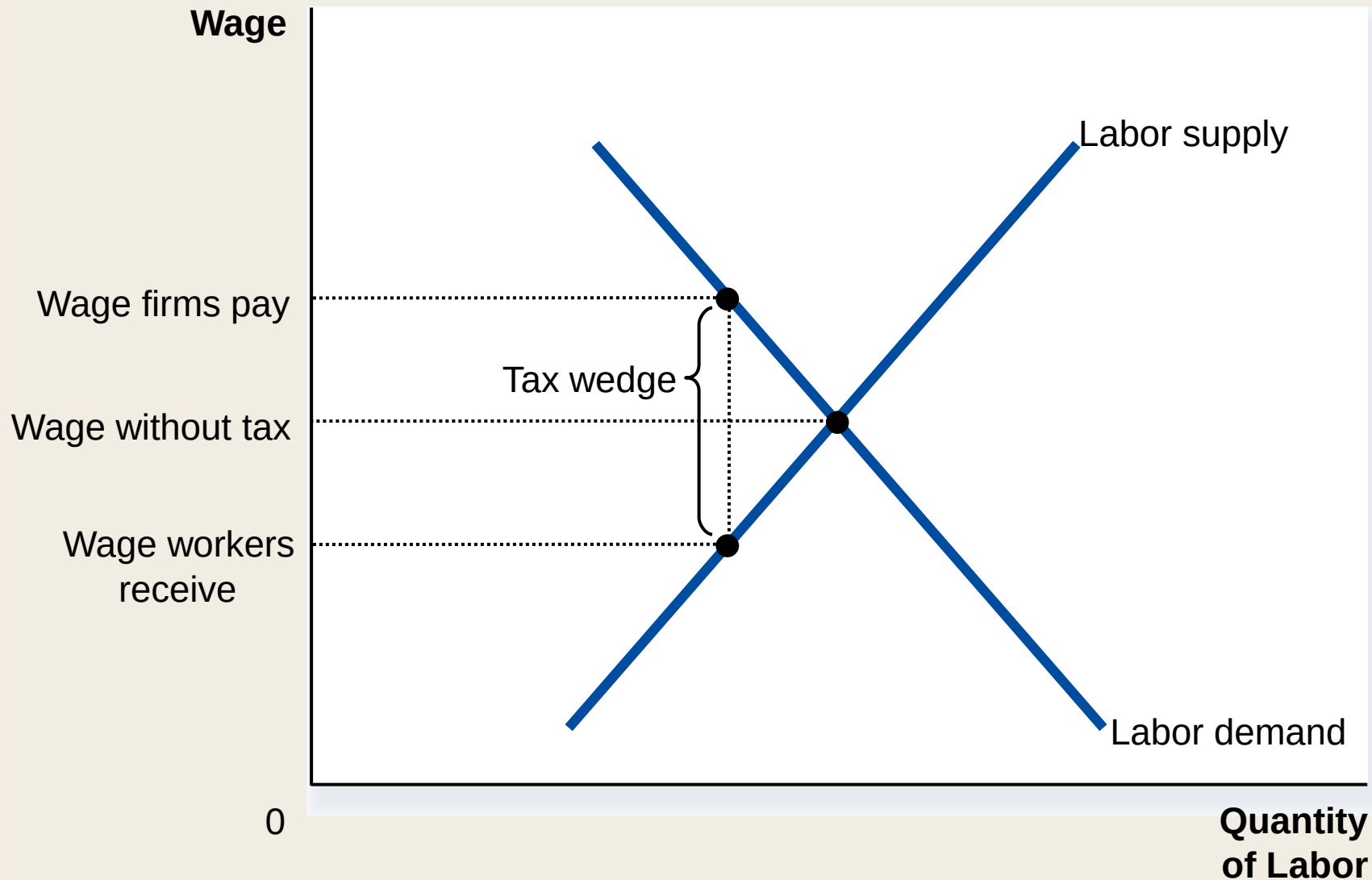


Elasticity and Tax Incidence

- What was the impact of tax?
 - Taxes discourage market activity.
 - When a good is taxed, the quantity sold is smaller.
 - Buyers and sellers share the tax burden.



Figure 8 A Payroll Tax



Elasticity and Tax Incidence

- In what proportions is the burden of the tax divided?
- How do the effects of taxes on sellers compare to those levied on buyers?
- The answers to these questions depend on the elasticity of demand and the elasticity of supply.

Figure 9 How the Burden of a Tax Is Divided

(a) Elastic Supply, Inelastic Demand

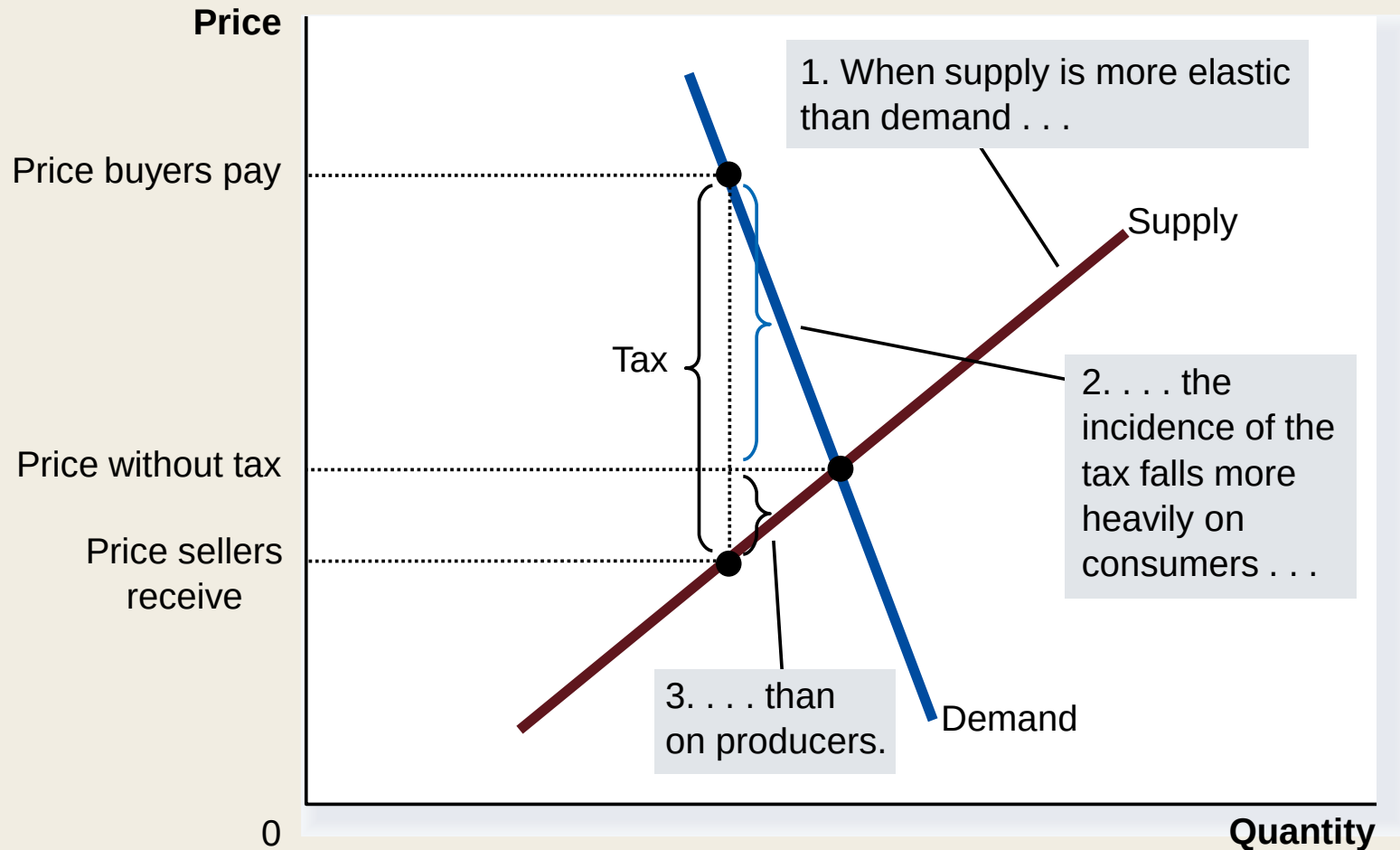
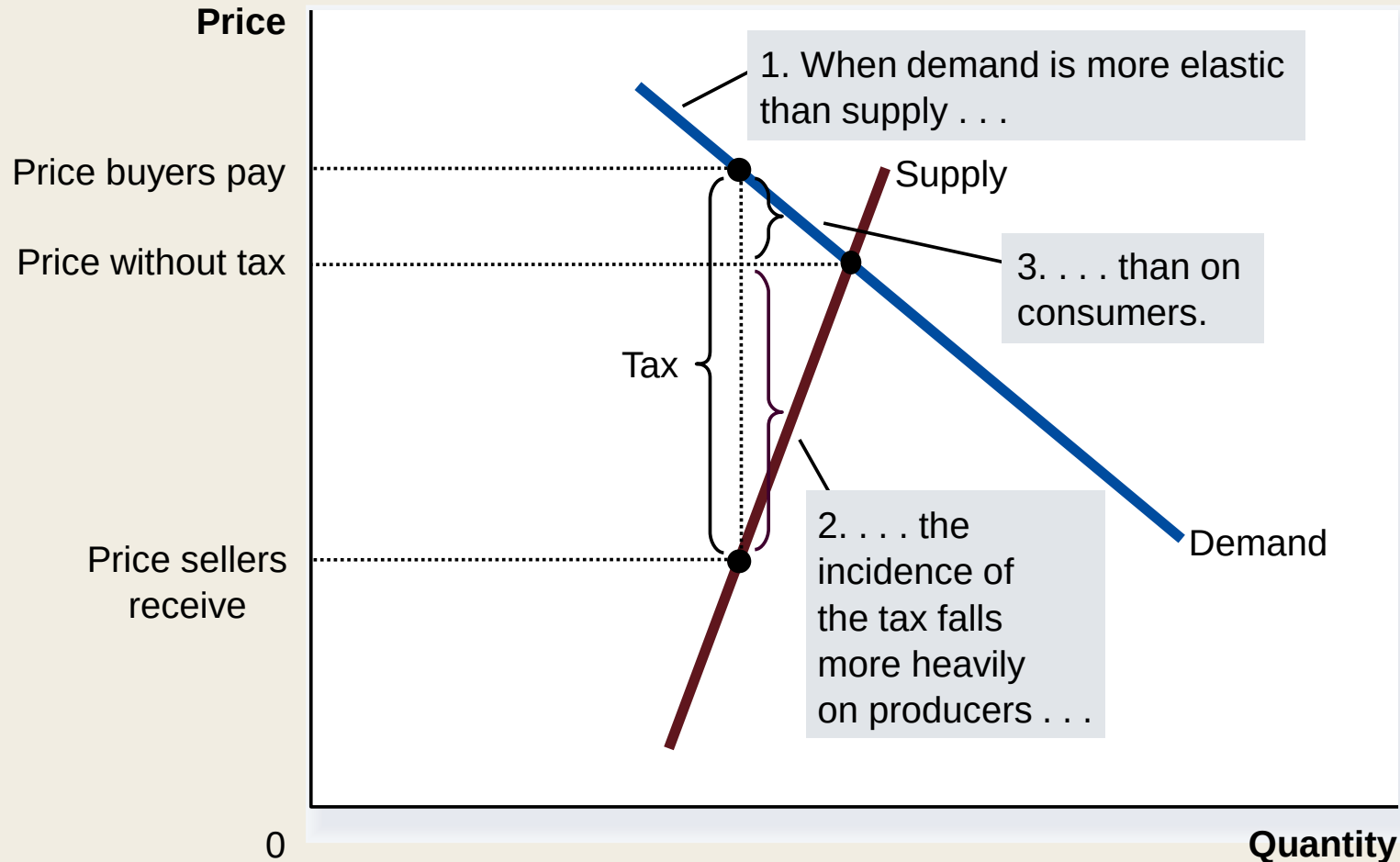


Figure 9 How the Burden of a Tax Is Divided

(b) Inelastic Supply, Elastic Demand



Elasticity and Tax Incidence

So, how is the burden of the tax divided?

The burden of a tax falls more heavily on the side of the market that is less elastic.



Summary

- Price controls include price ceilings and price floors.
- A price ceiling is a legal maximum on the price of a good or service.
 - An example is rent control.
- A price floor is a legal minimum on the price of a good or a service.
 - An example is the minimum wage.

Summary

- Taxes are used to raise revenue for public purposes.
- When the government levies a tax on a good, the equilibrium quantity of the good falls.
- A tax on a good places a wedge between the price paid by buyers and the price received by sellers.

Summary

- The incidence of a tax refers to who bears the burden of a tax.
- The incidence of a tax does not depend on whether the tax is levied on buyers or sellers.
- The incidence of the tax depends on the price elasticities of supply and demand.
- The burden tends to fall on the side of the market that is less elastic.